CursorVision

Kody Graham, Daniyar Alimkhanov, Seth Wojcik

Faculty Mentor(s): Md Jobair Hossain Faruk

University of Central Arkansas

Department of Computer Science

Abstract

CursorVision is a webcam-based, hands-free cursor control system for users with limited upper-limb mobility. The system uses a modified MediaPipe face mesh model to extract facial and eye landmarks, a TensorFlow-based intent prediction module to interpret movement, and a calibration process to improve precision. Cursor movement is generated through operating system APIs, and all processing is performed locally for privacy and low latency. Initial testing showed reliable real-time directional cursor control for basic hands-free navigation.

Implementation

- Built with **Python**
- Uses **OpenCV** for webcam capture and frame processing
- Uses **MediaPipe** for facial and eye landmark tracking
- Uses **TensorFlow** for intent prediction
- Uses OS cursor APIs for cursor control

Includes:

- A Calibration mode
- Real-time movement tracking
- Visual feedback/demo interface

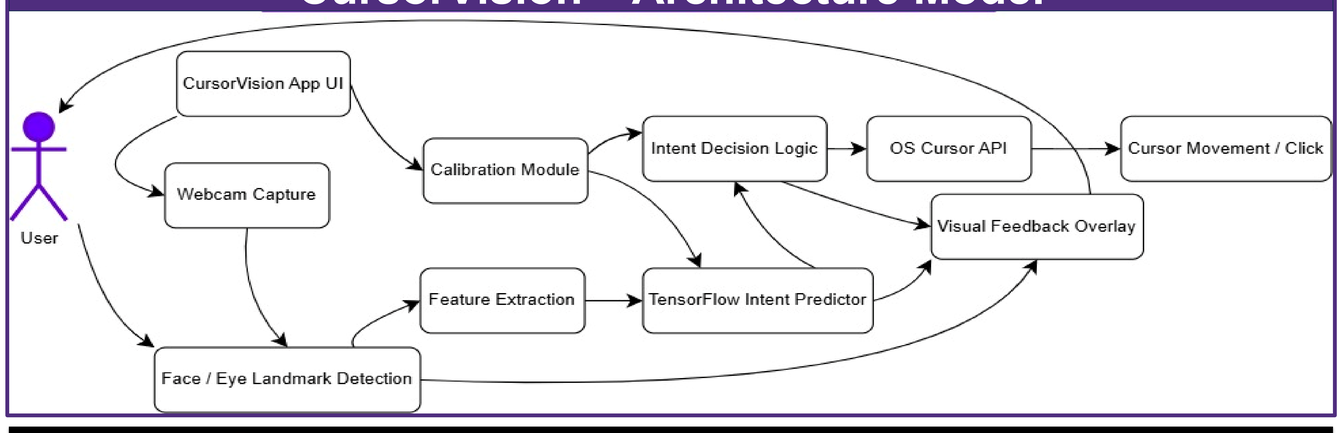
Problem Statement

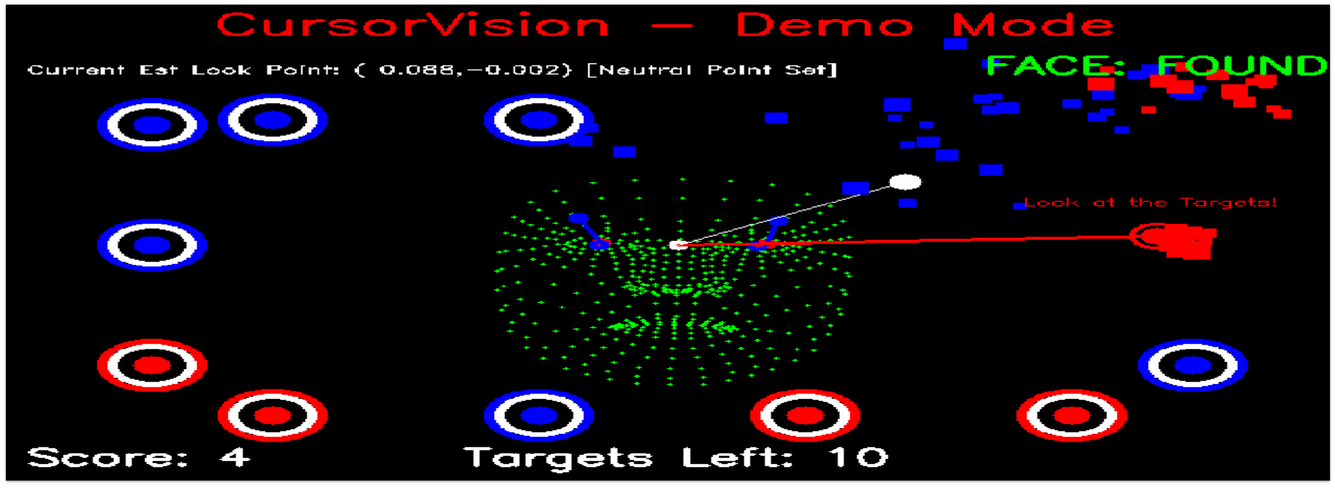
- Traditional mice and trackpads are difficult or unusable for some individuals with limited arm or hand mobility.
- Many assistive input systems require costly specialized hardware.
- A **low-cost**, webcam-based alternative could improve accessibility using standard consumer devices.
- CursorVision was developed to provide hands-free cursor control through facial and eye movement.

System Design

- Webcam captures live video input.
- Modified **MediaPipe** face mesh detects facial and eye landmarks in real time.
- Calibration module establishes a user-specific neutral position.
- Feature extraction and **TensorFlow** intent prediction estimate movement direction.
- Confidence filtering reduces noise and unstable cursor motion.
- **OS cursor API** converts predicted intent into cursor movement.
- System runs fully on-device for privacy and reduced latency.

CursorVision – Architecture Model





Results

- System achieved real-time facial and eye landmark tracking.
- Cursor movement responded consistently to directional user input.
- Calibration improved control precision and stability.
- Local execution reduced dependency on external hardware or cloud processing.
- Functional testing supported basic hands-free navigation tasks.

Conclusion

- CursorVision demonstrates that hands-free cursor control can be achieved using only a webcam and local software.
- The system provides a low-cost, non-invasive accessibility solution.
- Key contributions include real-time landmark tracking, intent prediction, calibration, and local cursor control.

Future Work

- Gesture-based interaction.
- Adaptive learning for personalized control.
- Improved stability and accuracy across users.

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